

pandas.DataFrame.max

df:

	A	B
0	1	4
1	2	5
2	3	6

`df.max()` →

max values:

A	3
B	6

dtype: int64

Contents

[What Makes This Ebook Unique](#)

[Tips for Effective Use of This Ebook](#)

Introduction

The "Axis" Parameter

The "skipna" Parameter

The "numeric_only" Parameter

The "**kwargs" Parameter

Author Biography

Other Ebooks

Contacts and Social Media

Sources & References

Thank You!

1st Edition

Why This E-book?

"The aim of this ebook is to give you the 'aha' moment right away at the start of learning a new concept."

- Practical step By Step Guide With Simple Examples
- Visual Illustrations and Interactive
- Simple Datasets
- Comprehensive Coverage (pandas Documentation used as reference)

Tips for Effective Use of This Ebook

Use Examples First: If you don't understand the text, go straight to the examples—they're self-explanatory. After reviewing them, return to the text to grasp the practical concepts.

Practice with Datasets: Copy the datasets provided and practice using them to reinforce your understanding.

Click on the blue links to go directly to the section you want to learn about.

Introduction

Return the maximum of the values over the requested axis.

df:

	A	B
0	1	4
1	2	5
2	3	6



max values:

A 3
B 6
dtype: int64

Code snippet:

df.max()

Syntax

```
DataFrame.max(axis=0, skipna=True, numeric_only=False, **kwargs)
```

The `axis` parameter

`axis=0` (or `axis='index'`)

The `max()` function is applied to each column.

For column 'A', the maximum value is 3.

For column 'B', the maximum value is 6.

df:

	A	B
0	1	4
1	2	5
2	3	6



Output:

```
A    3
B    6
dtype: int64
```

Code snippet:

```
df.max()
or
df.max(axis=0)
or
df.max(axis='index')
```

`axis=1` (or `axis='columns'`)

The `max()` function is applied to each row.

For row 0, the maximum value between 1 and 4 is 4.

For row 1, the maximum value between 2 and 5 is 5.

For row 2, the maximum value between 3 and 6 is 6.

df:

	A	B
0	1	4
1	2	5
2	3	6



Output:

```
0    4
1    5
2    6
dtype: int64
```

Code snippet:

```
df.max(axis=1)
or
df.max(axis='columns')
```

axis=None

The `max()` function is applied to all the values in the DataFrame.

It finds the overall maximum value, which is 6.

df:

	A	B
0	1	4
1	2	5
2	3	6



output:

6

Code snippet:

```
df.max(axis=None)
```

The `skipna` parameter

It excludes NA/null values when computing the result.

`skipna` is a boolean, defaulting to `True`

when `skipna=True` (Default)

The `max()` function is applied to each column.

In column 'A', the maximum value is 2 (ignoring the NaN).

In column 'B', the maximum value is 6 (ignoring the NaN).

df:

	A	B	C
0	1.0	4.0	7
1	2.0	NaN	8
2	NaN	6.0	9



Output:

A 2.0

B 6.0

C 9.0

dtype: float64

Code snippet:

```
df.max(skipna=True)
```

In column 'C', the maximum value is 9 (no NaN to ignore).

@Kichere The Data Scientist

when skipna=False

Includes NaN/None values in the calculation, resulting in NaN if any missing values are present.

The max() function is applied to each column.

In column 'A', because there is a NaN, the result is NaN.

In column 'B', because there is a NaN, the result is NaN.

In column 'C', the maximum value is 9 (since there is no NaN).

df:

	A	B	C
0	1.0	4.0	7
1	2.0	NaN	8
2	NaN	6.0	9



Output:

A NaN

B NaN

C 9.0

dtype: float64

Code snippet:

```
df.max(skipna=False)
```

The numeric_only parameter

It includes only float, int, boolean columns.

It is not implemented for Series.

It is boolean, defaulting to False

numeric_only=False (Default))

With `numeric_only=False`, all columns are included. The maximum of integers, floats, and booleans is computed. For strings, it returns the lexicographically maximum value.

df:

	A	B	C	D
0	1	4.0	True	x
1	2	5.5	False	y
2	3	6.1	True	z



Output:

```
A    3
B    6.1
C    True
D     z
dtype: object
```

Code snippet:

```
df.max()
or
df.max(numeric_only=False)
```

`numeric_only=True`

With `numeric_only=True`, only numeric columns (A, B, C) are considered. If all columns are non-numeric, the result will be `NaN`.

df:

	A	B	C	D
0	1	4.0	x	True
1	2	5.5	y	False
2	3	6.1	z	True



Output:

```
A    3
B    6.1
D    True
dtype: object
```

Code snippet:

```
df.max(numeric_only=True)
```

The `**kwargs` parameter

he `**kwargs` allows for the inclusion of additional keyword arguments that can be passed to the function, beyond the explicitly defined ones (`axis`, `skipna`, and `numeric_only`).

Sources & References

[pandas.Series.value_counts Documentation](https://pandas.pydata.org/docs/reference/api/pandas.Series.value_counts.html) (https://pandas.pydata.org/docs/reference/api/pandas.Series.value_counts.html)

Other E-books on pandas

[Download](https://github.com/kicherethedatascientist/My_e-Books) (https://github.com/kicherethedatascientist/My_e-Books)

Author Biography

Kichere Magubu is a data enthusiast and content creator.

Contacts and Social Media

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```
#print("The cell to convert jupyter notebook to html")  
!jupyter nbconvert --to hide_code_html "df.max.ipynb"
```